

SAIRA RIZVI

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RESEARCH INTERESTS

AI risk assessment and mitigation; trustworthy and safe AI systems; misinformation and information integrity; adversarial robustness of machine learning systems; applied machine learning for security and risk domains.

EDUCATION

University of Waterloo — Master of Mathematics, Data Science 2019 – 2021

Cumulative GPA: 85.80/100. Thesis: “Misinformation Retrieval,” supervised by Prof. Charles L. A. Clarke.

University of Toronto — Honours Bachelor of Science, Mathematics & Statistics 2015 – 2019

Conferred with Distinction. Cumulative GPA: 3.42/4.0. Achieved a 4.0/4.0 sessional GPA in third year (Winter 2018).

Recipient of an undergraduate scholarship in third year.

PUBLICATIONS

- Clarke, C. L. A., Maistro, M., Rizvi, S., Smucker, M. D., & Zuccon, G. (2020). Overview of the TREC 2020 Health Misinformation Track. In Proceedings of the Twenty-Ninth Text REtrieval Conference (TREC 2020), National Institute of Standards and Technology (NIST).

Served as a track coordinator for the TREC Health Misinformation Track, contributing to topic development, document collection curation, and evaluation methodology for a shared task addressing information retrieval in the presence of health misinformation.

PATENTS

- Systems & Methods for Phishing Campaign Detection, US20250337778A1 (Filed 2025)
- Detection of Phishing Websites Using Machine Learning, CA3170593A1 (Filed 2022)

RESEARCH & APPLIED EXPERIENCE

RBC — Security Insights & AI | Staff Data Scientist & Manager May 2024 – Present

Applied ML research and production systems for fraud and behavioral threat detection

- Built a detection system for coordinated political advocacy and harassment email campaigns targeting the CEO and other senior executives, modeling attacker-to-target structure as a complementary signal to content-based classification.
- Led development of a fraud detection model projected to capture \$10M CAD in fraud annually, from problem formulation through production deployment and evaluation.
- Designed a behavioral modeling approach for insider threat detection, investigating risk signals and validating the resulting system under strict false-positive constraints.
- Built a proof-of-concept biometric classification model, evaluating an emerging technique’s viability for identity-based risk detection.
- Led a cross-team research effort on spear-phishing detection, synthesizing findings for security and product stakeholders.

RBC — Security Insights & AI | Senior Data Scientist

Sept 2021 – May 2024

Anomaly detection for phishing and email-based threats

- Designed and deployed a phishing campaign detection system using anomaly detection, becoming the top-performing model on the team.
- Increased email threat detection coverage from ~20% to ~60% bank-wide through iterative feature and model research.
- Translated applied research into a patented phishing detection method.

University of Waterloo | Master’s Thesis & TREC Research

Sept 2019 – Sept 2021

Misinformation retrieval and detection

- Fine-tuned a BERT model (PyTorch, Hugging Face) for misinformation classification, covering data labeling, training, and evaluation.
- Contributed to the design and evaluation methodology of the TREC 2020 Health Misinformation Track as a track coordinator.

MENTORING

- Mentored junior data scientists at RBC on research scoping, experimental design, and evaluation practices.

HONORS & AWARDS

- RBC Individual Contribution Award (2026) – for high impact contributions over years in Global Cybersecurity Division at RBC
- RBC Performance Recognition (2025)
- RBC Team Performance Award (2024)
- 2nd Place, Best Cybersecurity Patent, RBC (2024)
- Honours B.Sc. Conferred with Distinction, University of Toronto (2019)

TECHNICAL SKILLS

- Languages & Frameworks: Python, SQL, PySpark, Scikit-learn, PyTorch, Hugging Face
- Methods: Anomaly detection, behavioral modeling, BERT/transformer fine-tuning, information retrieval, statistical modeling
- Infrastructure: AWS, Git, Jupyter, production ML pipelines

REFERENCES

Available upon request.